

STAT 525

# **Chapter 8**

## **Quantitative and Qualitative Predictors**

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## Polynomial Regression

- Multiple regression using  $X_i^2$ ,  $X_i^3$ , etc as additional predictors to fit/approximate a nonlinear regression
- Hierarchical approaching to fitting (model selection); usually no more than the third polynomial degree
- Generates quadratic, cubic polynomial relationships
- Can often lead to a multicollinearity problem
- Possible solution: centralize the predictors
  - Centering

$$\tilde{X}_{ij} = X_{ij} - \bar{X}_j,$$

where  $\bar{X}_j = \sum_i X_{ij}/n$

## Example: Power Cell (p.300)

- Response variable is the life (in cycles) of a power cell
- Explanatory variables are
  - Charge rate (3 levels)
  - Temperature (3 levels)
- This is a designed experiment
  - Notice  $\sum_i (X_{i1} - \bar{X}_1)(X_{i2} - \bar{X}_2) = 0 \rightarrow cor(X_1, X_2) = 0$
  - Notice  $cor(X_i, X_i^2)$  is a large value.
- Standardizing the explanatory variables
  - For example, we code the values of predictors as  $x_{ij} = 0, \pm 1$
  - Notice  $\sum x_{i1}x_{i1}^2 = 0 \rightarrow cor(x_1, x_1^2) = 0$
  - Notice  $\sum x_{i2}x_{i2}^2 = 0 \rightarrow cor(x_2, x_2^2) = 0$

```
options nocenter;
data a1;
    infile 'U:\.www\datasets525\Ch07ta09.txt';
    input cycles chrates temp;

proc print data=a1;
run;

data a1; set a1;
    chrates2=chrates*chrates;
    temp2=temp*temp;
    ct=chrates*temp;

proc reg data=a1;
    model cycles=chrates temp chrates2 temp2 ct;
run;
```

| Analysis of Variance |    |                |             |         |        |
|----------------------|----|----------------|-------------|---------|--------|
| Source               | DF | Sum of Squares | Mean Square | F Value | Pr > F |
| Model                | 5  | 55366          | 11073       | 10.57   | 0.0109 |
| Error                | 5  | 5240.43860     | 1048.08772  |         |        |
| Corrected Total      | 10 | 60606          |             |         |        |
| Root MSE             |    | 32.37418       | R-Square    | 0.9135  |        |
| Dependent Mean       |    | 172.00000      | Adj R-Sq    | 0.8271  |        |
| Coeff Var            |    | 18.82220       |             |         |        |

| Parameter Estimates |    |                    |                |         |         |
|---------------------|----|--------------------|----------------|---------|---------|
| Variable            | DF | Parameter Estimate | Standard Error | t Value | Pr >  t |
| Intercept           | 1  | 337.72149          | 149.96163      | 2.25    | 0.0741  |
| chrates             | 1  | -539.51754         | 268.86033      | -2.01   | 0.1011  |
| temp                | 1  | 8.91711            | 9.18249        | 0.97    | 0.3761  |
| chrates2            | 1  | 171.21711          | 127.12550      | 1.35    | 0.2359  |
| temp2               | 1  | -0.10605           | 0.20340        | -0.52   | 0.6244  |
| ct                  | 1  | 2.87500            | 4.04677        | 0.71    | 0.5092  |

- MULTICOLLINEARITY? Overall significant model but no individual variables significant

```
proc corr data=a1;
    var chrates temp chrates2 temp2 ct;
run; quit;
```

| Pearson Correlation Coefficients, N = 11 |         |         |          |         |         |
|------------------------------------------|---------|---------|----------|---------|---------|
| Prob >  r  under H0: Rho=0               |         |         |          |         |         |
|                                          | chrates | temp    | chrates2 | temp2   | ct      |
| chrates                                  | 1.00000 | 0.00000 | 0.99103  | 0.00000 | 0.60532 |
|                                          |         | 1.0000  | <.0001   | 1.0000  | 0.0485  |
| temp                                     | 0.00000 | 1.00000 | 0.00000  | 0.98609 | 0.75665 |
|                                          |         | 1.0000  | 1.0000   | <.0001  | 0.0070  |
| chrates2                                 | 0.99103 | 0.00000 | 1.00000  | 0.00592 | 0.59989 |
|                                          |         | <.0001  | 1.0000   | 0.9862  | 0.0511  |
| temp2                                    | 0.00000 | 0.98609 | 0.00592  | 1.00000 | 0.74613 |
|                                          |         | 1.0000  | <.0001   | 0.9862  | 0.0084  |
| ct                                       | 0.60532 | 0.75665 | 0.59989  | 0.74613 | 1.00000 |
|                                          |         | 0.0485  | 0.0070   | 0.0511  | 0.0084  |

- As anticipated, high correlation between  $X_1$  and  $X_1^2$  as well as  $X_2$  and  $X_2^2$

- SAS Codes: Standardizing

```
data a2; set a1;  
    schrate=chrate; stemp=temp;  
    keep cycles schrate stemp;  
  
proc standard data=a2 out=a3 mean=0 std=1;  
    var schrate stemp;  
  
proc print data=a3;  
run;  
  
data a3; set a3;  
    schrate2=schrate*schrate;  
    stemp2=stemp*stemp;  
    sct=schrate*stemp;  
  
proc reg data=a3;  
    model cycles=schrate stemp schrate2 stemp2 sct;  
run; quit;
```

- Note that this standardizes values so the mean is zero and variance 1. The key aspect is centering to remove the collinearity.

| Analysis of Variance |    |                |             |         |        |  |
|----------------------|----|----------------|-------------|---------|--------|--|
| Source               | DF | Sum of Squares | Mean Square | F Value | Pr > F |  |
| Model                | 5  | 55366          | 11073       | 10.57   | 0.0109 |  |
| Error                | 5  | 5240.43860     | 1048.08772  |         |        |  |
| Corrected Total      | 10 | 60606          |             |         |        |  |
| Root MSE             |    | 32.37418       | R-Square    | 0.9135  |        |  |
| Dependent Mean       |    | 172.00000      | Adj R-Sq    | 0.8271  |        |  |
| Coeff Var            |    | 18.82220       |             |         |        |  |

| Parameter Estimates |    |                    |                |         |         |
|---------------------|----|--------------------|----------------|---------|---------|
| Variable            | DF | Parameter Estimate | Standard Error | t Value | Pr >  t |
| Intercept           | 1  | 162.84211          | 16.60761       | 9.81    | 0.0002  |
| schrates            | 1  | -43.24831          | 10.23762       | -4.22   | 0.0083  |
| stemp               | 1  | 58.48205           | 10.23762       | 5.71    | 0.0023  |
| schrates2           | 1  | 16.43684           | 12.20405       | 1.35    | 0.2359  |
| stemp2              | 1  | -6.36316           | 12.20405       | -0.52   | 0.6244  |
| sct                 | 1  | 6.90000            | 9.71225        | 0.71    | 0.5092  |

## Remarks

- If we only care the final fitting with all polynomial terms, the two models are exactly the same. As a consequence, the overall  $F$  test/statistics are the same.
- But the individual coefficient estimator and inferences are completely different. After centralization, the two “main” effects are significant, implying that a linear model appears reasonable. Could do a general linear test (`test schrate2, stemp2, sct;`).
- Another possible way of handling the multicollinearity in polynomial regression is using orthogonal polynomial (STAT 514).

## Interaction Models

- With several explanatory variables, we need to consider the possibility that the effect of one variable depends on the value of another variable
- Model this relationship as the product of predictors
- Special Cases:
  - One binary (Y/N) and one continuous
  - Two continuous predictors

## Interaction Models: Special Case #1

- $X_1 = 0$  or 1 identifying two groups
- $X_2$  is a continuous variable

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \beta_3 X_{i1} X_{i2} + \varepsilon_i$$

- When  $X_1 = 0$  (Group 1)

$$Y_i = \beta_0 + \beta_2 X_{i2} + \varepsilon_i$$

- When  $X_1 = 1$  (Group 2)

$$Y_i = (\beta_0 + \beta_1) + (\beta_2 + \beta_3) X_{i2} + \varepsilon_i$$

- Results in two regression lines
- $\beta_2$  is the slope for Group 1
- $\beta_2 + \beta_3$  is the slope for Group 2
- Similar relationship for the intercepts
- Three Hypotheses of Interest
  - $H_0 : \beta_1 = \beta_3 = 0$  : regression lines are the same
  - $H_0 : \beta_1 = 0$  : intercepts are the same
  - $H_0 : \beta_3 = 0$  : slopes are the same

## Example: Insurance Innovation (p.316)

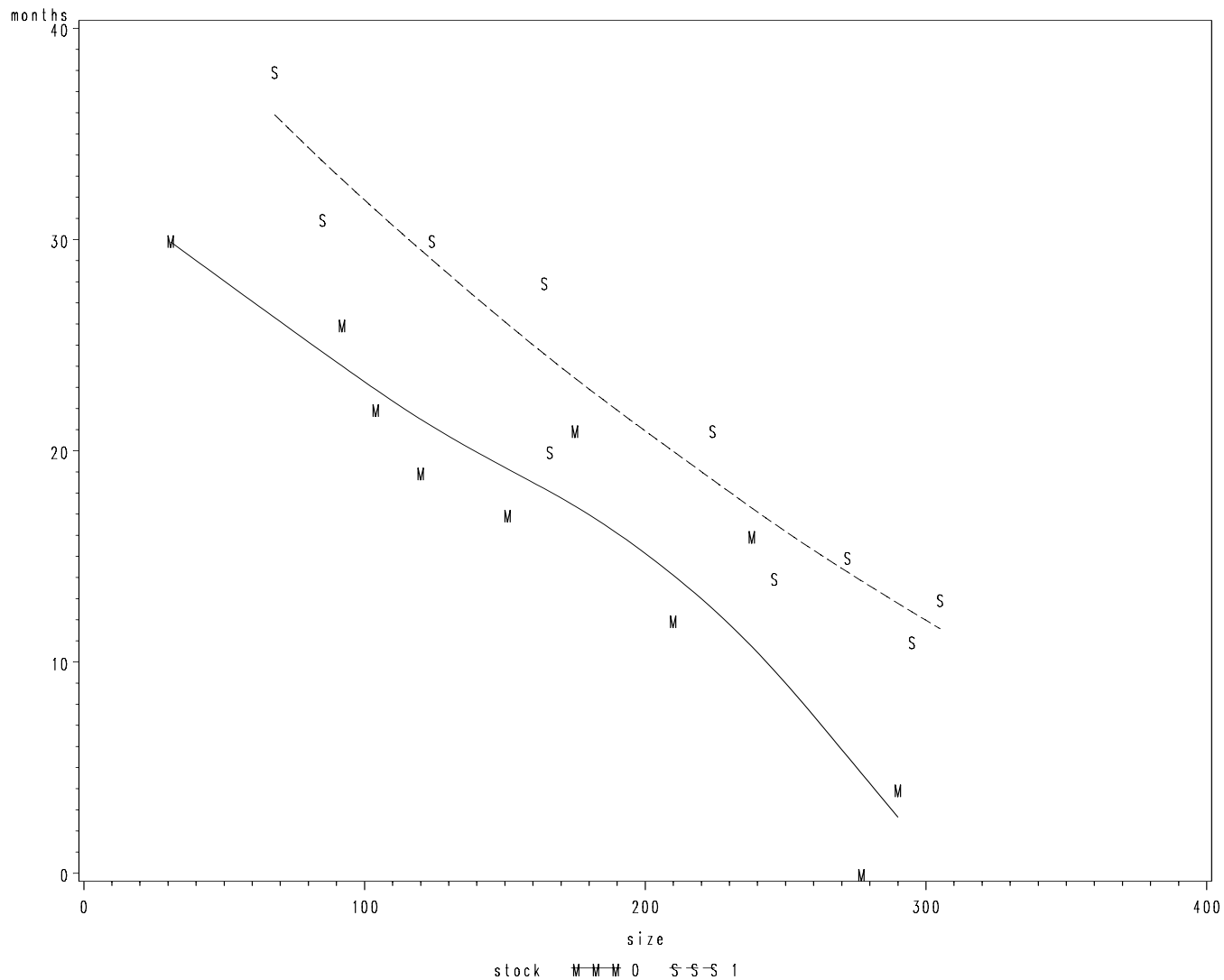
- $Y$  is the number of months for an insurance company to adopt an innovation
- $X_1$  is the size of the firm
- $X_2$  is the type of firm
  - $X_2 = 0 \rightarrow$  mutual fund firm
  - $X_2 = 1 \rightarrow$  stock firm
- Do stock firms adopt innovation faster? Is this true regardless of size?

```
data a1;  
  infile 'U:\.www\datasets525\Ch8ta02.txt';  
  input months size stock;
```

```

/* Scatterplot */
symbol1 v=M i=sm70 c=black l=1; symbol2 v=S i=sm70 c=black l=3;
proc sort data=a1; by stock size;
proc gplot data=a1;
    plot months*size=stock/frame;
run;

```



- Investigate the model:  $\text{months} = \text{size stock size*stock}$
- Test whether different types of firms adopt innovation at different paces regardless of size

```
data a1; set a1;  
    sizestoc=size*stock;  
  
proc reg data=a1;  
    model months=size stock sizestoc;  
    test stock, sizestock;  
run;
```

| Analysis of Variance |    |                |             |         |        |
|----------------------|----|----------------|-------------|---------|--------|
| Source               | DF | Sum of Squares | Mean Square | F Value | Pr > F |
| Model                | 3  | 1504.41904     | 501.47301   | 45.49   | <.0001 |
| Error                | 16 | 176.38096      | 11.02381    |         |        |
| Corrected Total      | 19 | 1680.80000     |             |         |        |
| Root MSE             |    | 3.32021        | R-Square    | 0.8951  |        |
| Dependent Mean       |    | 19.40000       | Adj R-Sq    | 0.8754  |        |
| Coeff Var            |    | 17.11450       |             |         |        |

| Parameter Estimates |    |                    |                |         |         |
|---------------------|----|--------------------|----------------|---------|---------|
| Variable            | DF | Parameter Estimate | Standard Error | t Value | Pr >  t |
| Intercept           | 1  | 33.83837           | 2.44065        | 13.86   | <.0001  |
| size                | 1  | -0.10153           | 0.01305        | -7.78   | <.0001  |
| stock               | 1  | 8.13125            | 3.65405        | 2.23    | 0.0408  |
| sizestoc            | 1  | -0.00041714        | 0.01833        | -0.02   | 0.9821  |

#### Test 1 Results for Dependent Variable months

| Source      | DF | Mean Square | F Value | Pr > F |
|-------------|----|-------------|---------|--------|
| Numerator   | 2  | 158.12584   | 14.34   | 0.0003 |
| Denominator | 16 | 11.02381    |         |        |

- Investigate the model: months = size stock

```
proc reg data=a1;
    model months=size stock;
run;
```

| Analysis of Variance |    |                |             |         |        |
|----------------------|----|----------------|-------------|---------|--------|
| Source               | DF | Sum of Squares | Mean Square | F Value | Pr > F |
| Model                | 2  | 1504.41333     | 752.20667   | 72.50   | <.0001 |
| Error                | 17 | 176.38667      | 10.37569    |         |        |
| Corrected Total      | 19 | 1680.80000     |             |         |        |
| Root MSE             |    | 3.22113        | R-Square    | 0.8951  |        |
| Dependent Mean       |    | 19.40000       | Adj R-Sq    | 0.8827  |        |
| Coeff Var            |    | 16.60377       |             |         |        |

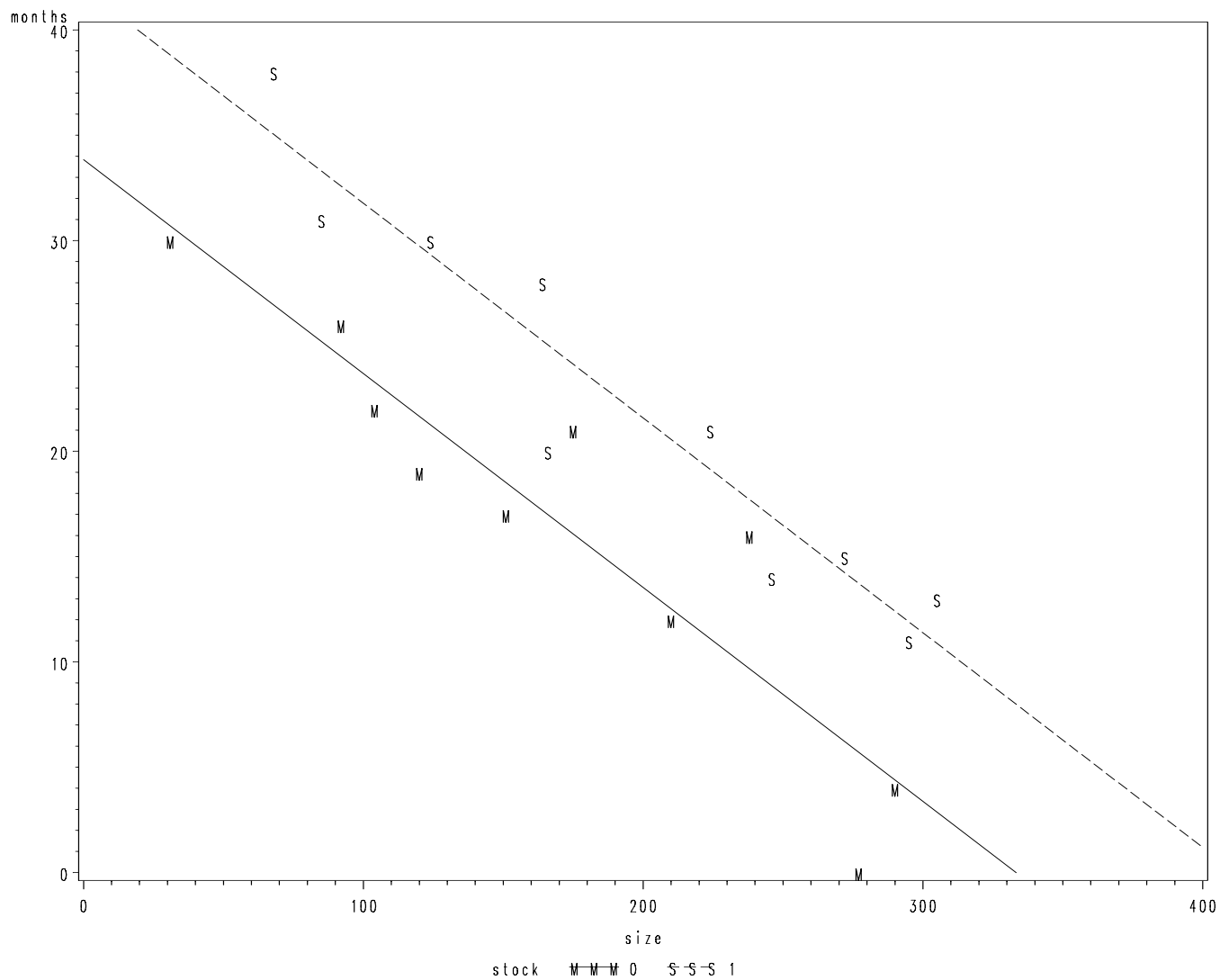
| Parameter Estimates |    |                    |                |         |         |
|---------------------|----|--------------------|----------------|---------|---------|
| Variable            | DF | Parameter Estimate | Standard Error | t Value | Pr >  t |
| Intercept           | 1  | 33.87407           | 1.81386        | 18.68   | <.0001  |
| size                | 1  | -0.10174           | 0.00889        | -11.44  | <.0001  |
| stock               | 1  | 8.05547            | 1.45911        | 5.52    | <.0001  |

- No apparent interaction here. The slope of the line does not depend on the type of firm.

```

/* Constant Slope */
symbol1 v=M i=r1 c=black;      symbol2 v=S i=r1 c=black;
proc gplot data=a1;
    plot months*size=stock/frame;
run; quit;

```



## Interaction Models: Special Case #2

- $X_1$  and  $X_2$  are continuous variables

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \beta_3 X_{i1} X_{i2} + \varepsilon_i$$

$$\beta_0 + \beta_1 X_1 + X_2 + \beta_3 X_1$$

- Can be written  $\beta_0 + (\beta_1 + \beta_3 X_2) X_1 + X_2$

$$\text{when } X_2 \text{ is fixed } Y_i = \beta_0 + (\beta_1 + \beta_3 X_{i2}) X_{i1} + \beta_2 X_{i2} + \varepsilon_i$$

$$\text{when } X_1 \text{ is fixed } Y_i = \beta_0 + \beta_1 X_{i1} + (\beta_2 + \beta_3 X_{i1}) X_{i2} + \varepsilon_i$$

- The coefficient of one explanatory variable depends on the value of the other explanatory variable
- Cannot discuss each predictor individually
- Use contour plot for visual interpretation:

## Contour plot in SAS

```
/* Contour plot for  $Y=10+X_1-3X_2+5X_1X_2$  */  
data a1;  
    do x1=0 to 1 by 0.1;  
        do x2=0 to 1 by 0.1;  
             $y=10+1*x_1-3x_2+5x_1*x_2$ ;  
            output;  
        end;  
    end;  
proc gcontour data=a1;  
    plot  $x_1*x_2=y$ ;  
run;
```

# Qualitative Predictors

*use two indicators to distinguish each*

## Coding a Variable with Two Classes

- The “Insurance Innovation” example:  $X_1$  = size of firm
- Either stock firm or mutual fund firm in the study  $\implies$  a qualitative predictor with two classes,

$$X_2 = \begin{cases} 1, & \text{if stock firm} \\ 0, & \text{otherwise} \end{cases} \quad X_3 = \begin{cases} 1, & \text{if mutual fund} \\ 0, & \text{otherwise} \end{cases}$$

- Cannot include both indicators in the model
- The model below contains perfectly collinear columns

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \beta_3 X_{i3} + \varepsilon_i$$

- SAS will drop the last column

- The additive model below is appropriate

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \varepsilon_i$$

- interpretation:

$$\begin{aligned} E\{Y_i\} &= \beta_0 + \beta_1 X_{i1} && \text{if mutual fund} \\ E\{Y_i\} &= (\beta_0 + \beta_2) + \beta_1 X_{i1} && \text{if stock firm} \end{aligned}$$

## Alternative Coding of a Two-Class Variable

*use one indicator to distinguish each*

- An alternative coding

$$X_2 = \begin{cases} 1 & , \text{ if stock firm} \\ -1 & , \text{ if mutual fund} \end{cases}$$

- The additive model is still appropriate

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \varepsilon_i$$

- interpretation:

$$\begin{aligned} E\{Y_i\} &= (\beta_0 + \beta_2) + \beta_1 X_{i1} && \text{if stock firm} \\ E\{Y_i\} &= (\beta_0 - \beta_2) + \beta_1 X_{i1} && \text{if mutual fund} \end{aligned}$$

- $\beta_0$  is an "average" intercept of regression line
- will use this coding for Analysis of Variance

## Coding a Variable with Three Classes

- use one indicator to distinguish each*  
• First option: extend the indicator

$$X_2 = \begin{cases} 0, & \text{if mutual fund} \\ 1, & \text{if stock firm} \\ 2, & \text{if foreign firm} \end{cases}$$

- The model below is still appropriate

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \varepsilon_i$$

- interpretation: enforces an equal change in  $E\{Y\}$  for each extra indicator

$$\begin{array}{ll} E\{Y_i\} = \beta_0 + \beta_1 X_{i1} & \text{if mutual fund} \\ E\{Y_i\} = (\beta_0 + \beta_2) + \beta_1 X_{i1} & \text{if stock firm} \\ E\{Y_i\} = (\beta_0 + 2\beta_2) + \beta_1 X_{i1} & \text{if foreign firm} \end{array}$$

- may be too restrictive

## Alternative Coding of a Three-Class Variable

- use two indicators to distinguish each
- Second option:

$$X_2 = \begin{cases} 1, & \text{if stock firm} \\ 0, & \text{otherwise} \end{cases} \quad X_3 = \begin{cases} 1, & \text{if foreign firm} \\ 0, & \text{otherwise} \end{cases}$$

↓  
dummy variable

|         |   | $X_2$ | $X_3$ |
|---------|---|-------|-------|
| mutual  | 1 | 0     | 0     |
| stock   | 2 | 1     | 0     |
| foreign | 3 | 0     | 1     |

- The model below contains two indicators

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \beta_3 X_{i3} + \varepsilon_i$$

company size

- interpretation:

$$\begin{aligned} E\{Y_i\} &= \beta_0 + \beta_1 X_{i1} && \text{if mutual fund} \\ E\{Y_i\} &= (\beta_0 + \beta_2) + \beta_1 X_{i1} && \text{if stock firm} \\ E\{Y_i\} &= (\beta_0 + \beta_3) + \beta_1 X_{i1} && \text{if foreign firm} \end{aligned}$$

Additive, no interaction

- also more flexibility in presence of interactions  $X_1X_2$  and  $X_1X_3$

Cross Product, Multiplication

# Constrained Regression

- At times may want to put constraints on regression coefficients

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon_i$$

- $\beta_1 = 5$

$$Y = \beta_0 + 5X_1 + \beta_2 X_2 + \varepsilon_i \quad \text{Another Simple Linear Regression}$$

- $\beta_1 = \beta_2$

$$Y = \beta_0 + \beta (X_1 + X_2) + \varepsilon_i \dots$$

*Reorganize...*

- Can do this in SAS by redefining explanatory variables
  - Page 268, redefine so model is  $\tilde{Y}$  vs  $X_2$
- Can also use **RESTRICT** statement in **PROC REG**

```
PROC REG DATA=test;
```

```
MODEL Y=X1 X2 X3;
```

```
RUN; QUIT;
```

- Restrict  $\beta_1 = 5$ : **RESTRICT X1=5;**

- Restrict  $\beta_1 = \beta_2$ : **RESTRICT X1=X2;**

## Chapter Review

- Polynomial Regression
- Interaction Models
- Qualitative Predictors
- Constrained Regression